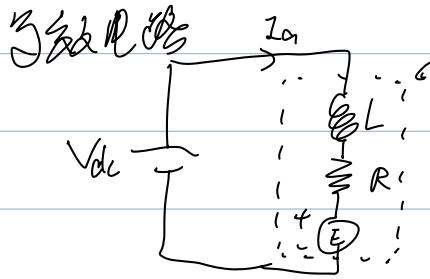
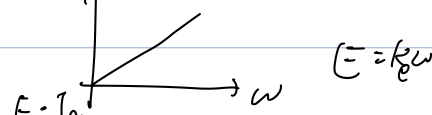
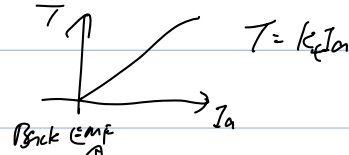


Brushed DC motor:

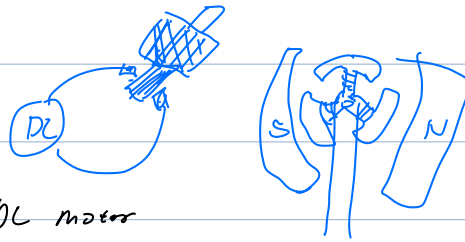


Brushed DC motor



Mechanical Power $P_m = E \cdot I_a$

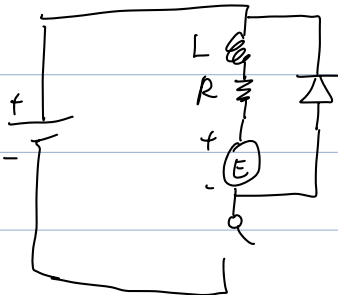
Proof: $E \cdot I_a = k_b \omega \cdot \frac{T}{k_f} = \frac{k_b}{k_f} \cdot \omega T = \omega T \quad (k_b = k_f)$



Brushed DC motor drive:

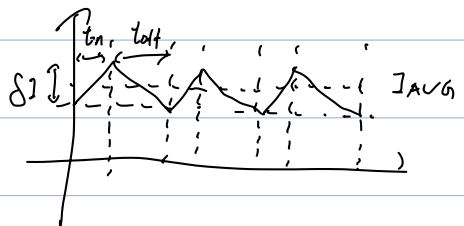
我們如何控制 drive 來調節速度:

- * Voltage control
- * Armature resistor control
- * Flux control



Switch on: $V_{dc} = L \left| \frac{di}{dt} \right| + I_{avg} R + E$

Switch off: $L \left| \frac{di}{dt} \right| = V_{diode} + I_{avg} R + E$

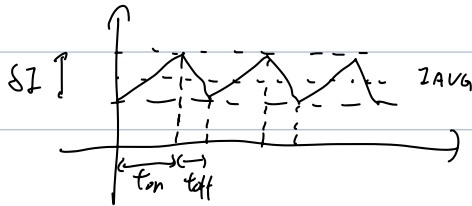


Example:

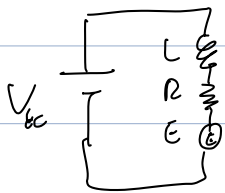
$$\text{PWM periode } T = \frac{1}{f_{\text{PWM}}} = \frac{1}{2\text{K}} = 0.5\text{ms}$$

$$\text{switch on time: } t_{\text{on}} = T \cdot \text{duty cycle} = 0.3\text{ms}$$

$$\text{switch off time: } t_{\text{off}} = T - t_{\text{on}} = 0.2\text{ms}$$



switch on:



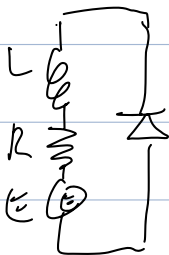
$$V_{dc} = L \left(\frac{\partial I}{\partial t_{\text{on}}} \right) + I_{\text{AVG}} R + E$$

$$\Rightarrow 12 = 0.5\text{m} \cdot \frac{\partial I}{0.3\text{m}} + I_{\text{AVG}} \cdot 70\text{m} + 2.2 \times \frac{2500}{1000}$$

$$\Rightarrow \partial I = 12 \times \frac{3}{5} - I_{\text{AVG}} \cdot 70\text{m} \times \frac{3}{5} - 2.2 \times 2.5 \times \frac{3}{5}$$

$$\Rightarrow \partial I = -0.042 \cdot I_{\text{AVG}} + 3.9 \quad (1)$$

switch off:



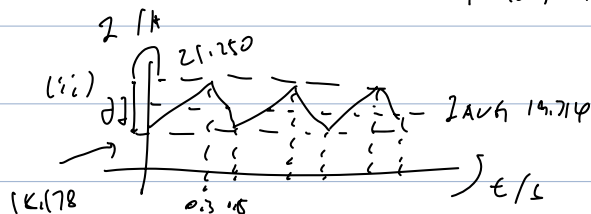
$$L \left(\frac{\partial I}{\partial t_{\text{off}}} \right) = I_{\text{AVG}} R + E + V_{\text{diode}}$$

$$\Rightarrow \frac{0.5}{0.2} \partial I = 70\text{m} \cdot I_{\text{AVG}} + 2.2 \times 2.5 + 0.8$$

$$\Rightarrow \partial I = 0.028 I_{\text{AVG}} + 2.52 \quad (2)$$

$$\text{From (1) (2)} \Rightarrow \partial I = 3.072 \text{ A}$$

$$I_{\text{AVG}} = 19.714 \text{ A}$$



$$(iii). P_{min} = I_{avg} = 5.5 \times 9.6 = 108 \text{ W}$$

$$= T \cdot \omega$$

$$\Rightarrow T = \frac{P_{min}}{\omega} = \frac{108}{2500 \times \frac{2\pi}{60}} = 0.4 \text{ / rev}$$